Experiment 1: Batch Processing Case Study – Student Marks Analysis

Aim

To implement batch processing using Python and Pandas for analyzing student marks, assigning grades, generating pass/fail status, and producing a summary of the dataset.

Theory

Batch processing is a data processing technique where a collection of records is processed together as a single batch instead of individually. It is commonly used when immediate processing is not required. In this experiment, all student records are loaded into memory, processed simultaneously, assigned grades and pass/fail status, and then saved into a new CSV file. Finally, statistical information such as average, highest, and lowest marks is generated.

Requirements

- Python 3.x
- Jupyter Notebook
- Pandas Library

Step 1: Import Library

```
import pandas as pd
```

Interpretation: Imports the Pandas library.

Step 2: Load Dataset

```
df = pd.read_csv("students_marks.csv")
print(df.head())
```

Interpretation: Loads the dataset.

Step 3: Create Grade Function

```
def calculate_grade(mark):
    ...
```

Interpretation: Defines grading criteria.

Step 4: Generate Grade and Status

```
df["Grade"]=df["Marks"].apply(calculate_grade)
df["Status"]=df["Marks"].apply(lambda x:"Pass" if x>=40 else "Fail")
```

Interpretation: Creates Grade and Status columns.

Step 5: Save Processed Dataset

```
df.to_csv("processed_data.csv",index=False)
```

Interpretation: Saves processed dataset.

Step 6: Batch Processing Summary

```
print(df["Marks"].mean())
print(df["Marks"].max())
```

```
print(df["Marks"].min())
```

Interpretation: Displays summary statistics.

Sample Input

```
Student,Marks
Rahul,95
Amit,82
Priya,76
Neha,68
Karan,54
Riya,39
Arjun,91
Anjali,73
Vikas,15
Sneha,94
```

Expected Output

```
Average Marks : 68.7
Highest Marks : 95
Lowest Marks : 15
```

Result

The batch processing system successfully processed the dataset and generated grades, status, and summary statistics.